

Out-of-Sample Evaluation of an Intraday Momentum Strategy for the S&P 500 ETF (SPY)

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Abstract

Zarattini, Aziz, and Barbon (2025) present an intraday momentum strategy for SPY that achieves an annualized return of 19.6% and a Sharpe Ratio of 1.33 over the period 2007–2024. This paper independently replicates their strategy using IQFeed 1-minute data and evaluates its performance on approximately two years of out-of-sample data (May 2024 to March 2026). Our in-sample replication closely matches the original results, confirming the reproducibility of the strategy (Sharpe Ratio 1.34 over 2015–2024). However, out-of-sample performance deteriorates substantially: the strategy achieves a total return of only 9.4% versus 29.5% for a passive buy-and-hold, with a Sharpe Ratio of 0.39. The decline in risk-adjusted returns is statistically significant. We examine potential explanations for this degradation, including changing market microstructure, the proliferation of zero-days-to-expiration (0-DTE) options, and the possibility that the observed underperformance falls within historical drawdown norms. Our findings highlight the importance of out-of-sample validation for intraday trading strategies and contribute to the ongoing discussion about the persistence of momentum anomalies at high frequencies.

Keywords: Intraday Momentum, Out-of-Sample Testing, Day Trading, Trend-Following, SPY, Market Microstructure

JEL Classification: G11, G12, G14

1 Introduction

Momentum strategies, first documented by [Jegadeesh and Titman \(1993\)](#), remain among the most studied anomalies in financial markets. While the majority of academic research focuses on cross-sectional momentum over monthly or quarterly horizons, a growing body of work examines time-series momentum at intraday frequencies ([Gao et al., 2018](#); [Baltussen et al., 2021](#); [Rosa, 2022](#); [Zarattini and Aziz, 2024](#)). [Rosa \(2022\)](#), in particular, documents a post-2013 decline in the predictability of intraday momentum, attributing it to market adaptation—a theme directly relevant to the out-of-sample question we examine here.

[Zarattini et al. \(2025\)](#) propose an intraday momentum strategy for SPY that identifies demand/supply imbalances using a time-varying “Noise Area” derived from recent price action. When the market moves outside this equilibrium zone, the strategy initiates trend-following positions with dynamic trailing stops incorporating the Volume Weighted Average Price (VWAP). Combined with volatility-targeting position sizing, the authors report a total return of 1,985% from 2007 to 2024, with a Sharpe Ratio of 1.33 and near-zero correlation with the broader market.

These results are compelling, but they are evaluated exclusively in-sample. The strategy’s parameters—the 14-day lookback for the Noise Area, the 30-minute trade frequency, the VWAP-based trailing stops, and the 2% volatility target—are all fixed choices that, while not formally optimized, were presumably selected because they produced favorable results on the historical data. Without out-of-sample validation, it is impossible to distinguish genuine market inefficiency from favorable parameter selection.

This paper addresses that gap. We independently replicate the [Zarattini et al. \(2025\)](#) strategy and test it on nearly two years of data that postdate their study. Our contribution is straightforward: we provide the first independent out-of-sample evaluation of this widely discussed intraday momentum approach.

2 Data

We use 1-minute OHLCV data for SPY sourced from IQFeed (DTN), covering the period from January 2, 2015 to March 31, 2026. The dataset comprises 1,101,234 intraday bars across 2,827 trading days. All bars are filtered to regular market hours (09:30–15:59 Eastern Time).

Our data begins in 2015 rather than 2007, so our in-sample period (2015 to April 2024) is shorter than the original paper’s. However, this does not affect the out-of-sample evaluation, which is the primary contribution of this study. The in-sample period serves only to confirm that our implementation faithfully reproduces the original strategy’s behavior.

We do not adjust for SPY dividends, which are paid quarterly and represent approximately 1.3% annually relative to SPY’s price level during our sample period. The implementation of [Zarattini et al. \(2025\)](#) includes a dividend adjustment for the previous-day close used in the Noise Area calculation. We estimate this omission has a negligible impact on results.

We define two evaluation periods:

- **In-sample (IS):** January 2, 2015 to April 30, 2024 (2,333 trading days)
- **Out-of-sample (OOS):** May 1, 2024 to March 31, 2026 (480 trading days)

The OOS start date corresponds to the end of the original paper’s data period.

3 Strategy Description

We implement the strategy exactly as described by [Zarattini et al. \(2025\)](#). The following restates their method for completeness; the strategy specification is entirely theirs. All parameter choices, signal rules, and risk-management components originate from the original paper.

3.1 Noise Area

For each trading day t and time-of-day $HH:MM$, a “Noise Area” represents the expected range of price movement under balanced buying and selling pressure. The boundaries are computed as follows:

1. For each of the prior 14 trading days, compute the absolute percentage move from the open to the close at time $HH:MM$.
2. Average these 14 values to obtain $\bar{\sigma}$.
3. Define the boundaries:

$$UB_t = \max(\text{Open}_t, \text{Close}_{t-1}) \times (1 + \bar{\sigma}) \quad (1)$$

$$LB_t = \min(\text{Open}_t, \text{Close}_{t-1}) \times (1 - \bar{\sigma}) \quad (2)$$

The use of max/min with the previous close accounts for overnight gaps.

3.2 Signal Generation

At 30-minute intervals (10:00, 10:30, 11:00, ...), the strategy evaluates whether the current close price is:

- Above the Upper Boundary **and** above the VWAP $\rightarrow$ Long signal (+1)
- Below the Lower Boundary **and** below the VWAP $\rightarrow$ Short signal (−1)
- Otherwise $\rightarrow$ No signal (0)

Positions are held until the next signal change or market close. Because a long position requires the price to be above *both* the Upper Boundary and the VWAP (and a short below both), a position is sustained only while the price remains beyond the more restrictive of the two levels.

3.3 Trailing Stops

In the specification of [Zarattini et al. \(2025\)](#), positions are opened on a breach of the Noise Area boundary and closed by a trailing stop set at the maximum of the boundary and the VWAP, drawing on the VWAP trend-trading approach of [Zarattini and Aziz \(2023\)](#):

- Long positions: $\text{stop} = \max(\text{UB}, \text{VWAP})$
- Short positions: $\text{stop} = \min(\text{LB}, \text{VWAP})$

Our implementation encodes this exit directly within the signal condition of Section 3.2. At the half-hourly decision points, requiring the close to exceed both the Upper Boundary and the VWAP is equivalent to entering on the boundary breach and holding the long until the price falls below $\max(\text{UB}, \text{VWAP})$; the short side is symmetric. We therefore evaluate a single combined condition rather than a separate entry rule and stop, but the resulting position path is the same. This trailing-stop design is the key innovation of [Zarattini et al. \(2025\)](#), as it significantly reduces drawdowns compared to using the opposite boundary as a stop.

3.4 Position Sizing

Position size targets a daily portfolio volatility of 2%, using the 14-day standard deviation of SPY daily returns ($\hat{\sigma}_{\text{SPY}}$):

$$\text{Shares} = \frac{\text{AUM}}{\text{Open}_t} \times \min\left(\frac{0.02}{\hat{\sigma}_{\text{SPY}}}, 4\right) \quad (3)$$

Following mainstream brokerage limits, the strategy caps leverage at four times capital ([Zarattini et al., 2025](#)).

3.5 Transaction Costs

We apply a commission of \$0.0035 per share and a slippage of \$0.001 per share, consistent with the assumptions of [Zarattini et al. \(2025\)](#), based on Interactive Brokers pricing.

4 Results

4.1 In-Sample Replication

Table 1 compares our in-sample replication with the results reported by [Zarattini et al. \(2025\)](#).

Table 1: In-Sample Replication vs. Original Paper

Metric	Paper (2007–2024)	Replication (2015–2024)
Total Return (%)	1,985	432.6
Annualized Return (%)	19.6	19.8
Annualized Volatility (%)	14.3	14.3
Sharpe Ratio	1.33	1.34
Hit Ratio (%)	43	44.8
Maximum Drawdown (%)	25	26.2

The annualized return (19.8% vs. 19.6%), volatility (14.3% vs. 14.3%), and Sharpe Ratio (1.34 vs. 1.33) match the original study almost exactly. The lower total return reflects our shorter sample period (9.3 vs. 17 years). The in-sample daily mean return of 0.076% is highly statistically significant ($t = 4.06$, $p < 0.001$). We conclude that our implementation faithfully reproduces the original strategy.

4.2 Out-of-Sample Performance

Table 2 presents the out-of-sample results alongside the in-sample replication and SPY buy-and-hold.

Table 2: In-Sample vs. Out-of-Sample Performance

Metric	IS (2015–2024)	OOS (May 2024–Mar 2026)
Total Return (%)	432.6	9.4
Annualized Return (%)	19.8	4.8
Annualized Volatility (%)	14.3	14.7
Sharpe Ratio	1.34	0.39
Hit Ratio (%)	44.8	45.0
Max Drawdown (%)	26.2	18.5
Best Day (%)	9.12	9.55
Worst Day (%)	−5.15	−3.10
SPY Buy & Hold Total (%)	143.6	29.5
SPY Buy & Hold Sharpe	0.62	0.91

Out-of-sample, the strategy’s annualized return drops from 19.8% to 4.8%, and the Sharpe Ratio declines from 1.34 to 0.39. The strategy returns 9.4% total while SPY buy-and-hold returns 29.5% over the same period. The OOS daily mean return of 0.023% is not statistically different from zero ($t = 0.54$, $p = 0.59$). The strategy does not generate significant alpha over SPY in the OOS period (excess daily return = -0.036% , $t = -0.54$, $p = 0.59$).

Notably, the hit ratio remains stable at approximately 45% in both periods, and the volatility profile is unchanged (14.3% IS vs. 14.7% OOS). The deterioration is entirely in the magnitude of winning trades relative to losing trades, not in the frequency of wins.

4.3 Year-by-Year Breakdown

Table 3 presents the annual performance of the strategy alongside SPY buy-and-hold.

Table 3: Year-by-Year Performance

Year	Strategy (%)	SPY B&H (%)	Sharpe	Hit (%)	Period
2015	21.0	−1.0	1.64	46.9	IS
2016	−13.4	9.6	−0.86	34.0	IS
2017	−8.8	19.4	−0.73	35.1	IS
2018	60.5	−6.3	2.70	43.9	IS
2019	8.4	28.7	0.77	42.0	IS
2020	26.2	16.1	1.60	44.7	IS
2021	32.8	27.0	1.97	57.6	IS
2022	23.7	−19.5	1.55	45.9	IS
2023	37.8	24.3	2.56	52.5	IS
2024	30.1	23.3	1.91	46.5	Split
2025	−2.7	16.4	−0.10	44.4	OOS
2026	−3.1	−4.6	−0.99	46.2	OOS

The yearly data reveals that the OOS underperformance is not unprecedented. The strategy experienced two consecutive negative years during the in-sample period (2016: −13.4%, 2017: −8.8%), a drawdown of similar magnitude to the current OOS period. However, the 2016–2017 drawdown occurred during a period of historically low volatility and strong trending markets that favored buy-and-hold, conditions that also characterize much of 2025.

4.4 Statistical Tests

We conduct four formal statistical tests to evaluate the significance of the OOS deterioration.

Table 4: Statistical Tests

Test	Statistic	p -value	Result
IS mean return > 0	4.064	< 0.001	Significant
OOS mean return > 0	0.542	0.588	Not significant
IS mean $\neq$ OOS mean (Welch’s t)	1.139	0.255	No significant difference
Sharpe IS $\neq$ Sharpe OOS	17.053	< 0.001	Significant decline

The results present a nuanced picture. While the in-sample mean return is highly significant and the OOS mean return is not, a direct comparison of the two means does not reject equality at conventional significance levels ($p = 0.255$). This is partly because

480 OOS days provide limited statistical power to detect a difference from a relatively noisy daily return series.

However, the decline in Sharpe Ratio from 1.34 to 0.39 is highly statistically significant ($z = 17.05$, $p < 0.001$), using the standard error approximation from Lo (2002). This suggests the change in risk-adjusted performance is unlikely to be explained by sampling variation alone.

4.5 Rolling Performance Analysis

Figure 1 displays the rolling 12-month Sharpe Ratio over the full sample period. The strategy maintained a rolling Sharpe above 1.0 for most of the period from mid-2018 through mid-2024, with peaks exceeding 3.0 during 2019 and late 2023. After the OOS boundary (May 2024), the rolling Sharpe declines sharply, turning negative by early 2026.



Figure 1: Rolling 12-month Sharpe Ratio. The vertical dashed line marks the OOS boundary (May 2024). The strategy’s rolling Sharpe declines from above 2.0 to below zero after the OOS boundary.

This pattern is consistent with a structural change in strategy profitability rather than normal cyclical variation. The 2016–2017 drawdown, while similar in depth, was followed by an immediate recovery. The current decline shows no signs of reversal through March 2026.

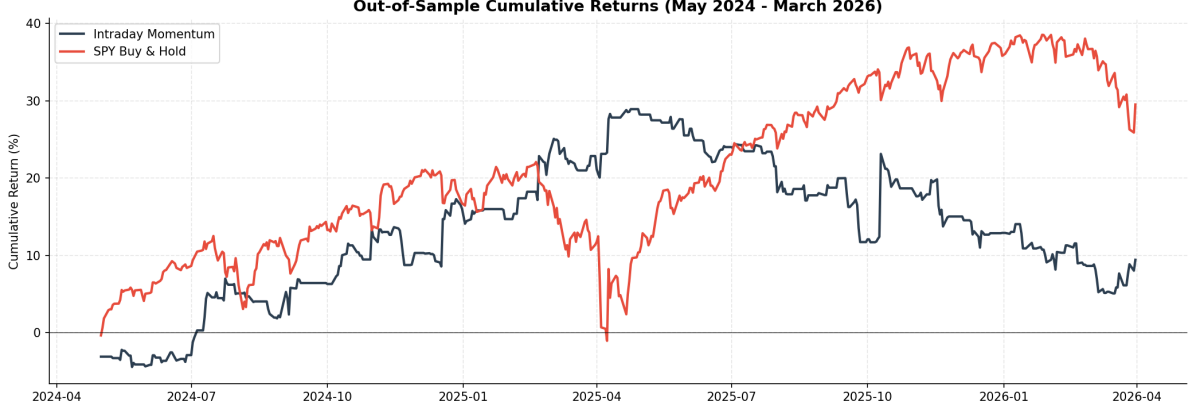


Figure 2: Cumulative equity curve: strategy (blue) vs. SPY buy-and-hold (gray). The vertical dashed line marks the OOS boundary. Out-of-sample, the strategy underperforms buy-and-hold by approximately 20 percentage points.

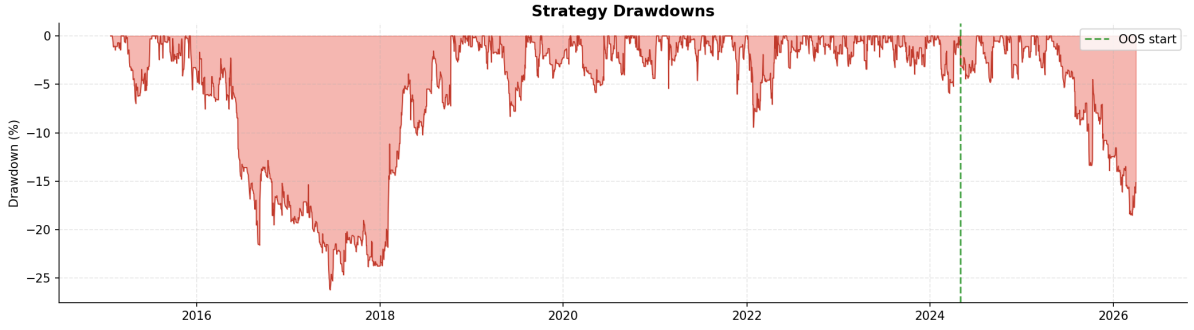


Figure 3: Drawdown profile over the full sample period. The OOS period shows a persistent drawdown without the sharp recoveries observed during in-sample drawdowns (e.g., 2018, 2020).

5 Discussion

5.1 Potential Explanations

Several factors may contribute to the OOS performance degradation:

Market microstructure changes. The explosive growth of 0-DTE (zero days to expiration) options on SPY since 2022 has fundamentally altered intraday price dynamics. Daily 0-DTE volume now regularly exceeds that of longer-dated options, and the associated gamma hedging by dealers creates intraday mean-reversion pressure that directly opposes the trend-following logic of the strategy. [Zarattini et al. \(2025\)](#) acknowledge this mechanism in Section 4.5 of their paper but do not test for its impact on strategy profitability over time.

Crowding. The original paper has received significant attention in the active trading community since its initial publication in 2024. Increased adoption of similar strategies could reduce available profits through crowding effects, particularly in a highly liquid instrument like SPY where marginal profits per trade are small.

Regime dependence. The strategy performs best in volatile, trending markets (2018, 2020, 2022) and worst in low-volatility, range-bound or steadily trending markets (2016–2017, 2025). The bull market of 2024–2025 was characterized by relatively steady upward momentum with few sharp dislocations, conditions that generate signals but produce small follow-through.

Normal variation. With only 480 OOS trading days, the evidence against the strategy is suggestive but not conclusive. The Welch t -test fails to reject equality of means ($p = 0.255$), and the strategy’s prior history includes drawdowns of comparable magnitude. A definitive assessment may require additional years of OOS data.

5.2 Limitations

Our study has several limitations. First, our in-sample period begins in 2015, missing the 2007–2014 period that includes the 2008 financial crisis. The strategy likely performed well during that volatile period, which could bias our in-sample Sharpe Ratio estimate. Second, we omit dividend adjustments, which may cause minor discrepancies in the Noise Area calculation around ex-dividend dates. Third, our OOS period of approximately two years, while meaningful, is short by the standards of long-run strategy evaluation. Finally, we do not replicate the original paper’s Matlab implementation; differences in floating-point arithmetic, data alignment, or edge-case handling could produce minor deviations.

6 Conclusion

We independently replicate the intraday momentum strategy proposed by [Zarattini et al. \(2025\)](#) and evaluate it on nearly two years of out-of-sample data. Our in-sample results closely match the original paper’s reported performance, confirming the reproducibility of their findings.

Out-of-sample, the strategy’s performance degrades substantially. The Sharpe Ratio declines from 1.34 to 0.39, the annualized return drops from 19.8% to 4.8%, and the strategy underperforms SPY buy-and-hold by approximately 20 percentage points. Statistical tests confirm the decline in risk-adjusted returns is significant, though the limited OOS sample size prevents definitive conclusions about whether the underlying edge has disappeared.

Our findings do not invalidate the original paper’s contribution. The strategy produced genuine risk-adjusted returns over a long in-sample period, and the observed OOS decline is consistent with known patterns in quantitative strategy lifecycle: initial discovery, publication, crowding, and decay. Whether the current underperformance represents a temporary drawdown or permanent impairment remains an open question that additional years of data will help resolve.

This work underscores the importance of continuous out-of-sample monitoring for intraday trading strategies. We encourage future researchers to extend this analysis as additional data becomes available.

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A Data and Code Availability

All backtests were conducted using Python 3.11 with 1-minute OHLCV data from IQFeed (DTN). The full replication code, including the strategy implementation, statistical tests, and figure-generation scripts, is publicly available at <https://github.com/PazSheimy/>

`spy-intraday-momentum-oos`. The underlying 1-minute IQFeed data is licensed and therefore not redistributed; the repository documents the required data format so the results can be reproduced with an equivalent 1-minute source.